

Physics 4700H: Assignment II

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Problem 1. Using notation appropriate to a continuous variable, show that the variance $\langle(\Delta x)^2\rangle = \langle x^2\rangle - \langle x\rangle^2$, where $\Delta x \equiv x - \langle x\rangle$.

Solution 1. *Proof.* In the continuous case

$$\begin{aligned}\langle x^2 \rangle &= \int x^2 p(x) dx, \\ \langle x \rangle^2 &= \left(\int x p(x) dx \right)^2, \\ \langle (\Delta x)^2 \rangle &= \int (\Delta x)^2 p(x) dx \\ &= \int (x - \langle x \rangle)^2 p(x) dx && \text{Definition} \\ &= \int (x^2 - 2x\langle x \rangle + \langle x \rangle^2) p(x) dx && \text{Binomial expansion} \\ &= \langle x^2 \rangle + \int \langle x \rangle^2 p(x) - 2x\langle x \rangle p(x) dx && \text{Definition of } \langle x^2 \rangle \\ &= \langle x^2 \rangle + \langle x \rangle^2 - 2\langle x \rangle \int x p(x) dx && \langle x \rangle \text{ is a constant, } \int p(x) dx = 1 \\ &= \langle x^2 \rangle + \langle x \rangle^2 - 2\langle x \rangle^2 && \text{Definition of } \langle x \rangle \\ &= \langle x^2 \rangle - \langle x \rangle^2 && \square\end{aligned}$$

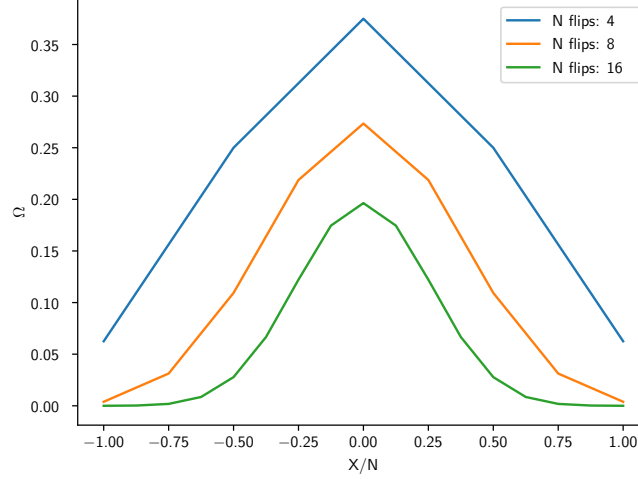
Problem 2. Consider the case of N tosses of a fair coin. The multiplicity function describes the number of ways a value of $X = N_H - N_T$ can be obtained,

$$\Omega(N, X) = \frac{N!}{\left[\frac{1}{2}(N+X)\right]! \left[\frac{1}{2}(N-X)\right]!}$$

- (a) Make a plot of $\Omega(N, X)/2^N$ versus X/N for three system sizes: $N = 4, 8, 16$. (Note that these distributions are discrete, not continuous.) Describe in words how the shape of the graph changes as N increases. You may construct the plot by hand (in which case show your calculations) or using a computer (in which case include your code).
- (b) Use the stronger form of Stirling's approximation to write an expression for $\ln \Omega(N, X)$ which is true when $N \gg 1$ and $|X|/N \ll 1$ (so $N \pm X$ also $\gg 1$).
- (c) Calculate the Taylor expansion of $\ln(1+x)$ for small x to second order.
- (d) Use your results in (b) and (c) to show that $\ln \Omega \approx N \ln(2) - X^2/(2N)$.
- (e) Using (d), find an approximate expression for Ω .

Solution 2.

- (a) The tails of the distribution flatten out as more probability “moves” towards the center and the lineshape approaches a Gaussian.



```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.special import factorial # Used here over math.factorial as it can handle element-wise factorials of
  ↪ arrays
4
5
6 def Omega(N: int, X: int) -> int:
7     return factorial(N) / (factorial(0.5 * (N + X)) * factorial(0.5 * (N - X)))
8
9
10 Ns = [4, 8, 16]
11 for N in Ns:
12     X_vals = np.array([x for x in range(-N, N + 1, 2)])
13     v = Omega(N, X_vals) / 2**N
14     plt.plot(X_vals / N, v, label=f"N flips: {N}")
15
16 plt.xlabel("X/N")
17 plt.ylabel(r"$\Omega$")
18 plt.legend()
19 plt.show()

```

- (b) The strong form of Stirling's approximation is, without the extra constants which here are negligible compared to N ,

$$\ln(N!) \approx N \ln(N) - N$$

so for our expression

$$\ln(\Omega(N, X)) = \ln\left(\frac{N!}{\left[\frac{1}{2}(N+X)\right]! \left[\frac{1}{2}(N-X)\right]!}\right).$$

Splitting this up with properties of $\ln$ we obtain three terms,

$$\ln(N!) - \ln\left(\left[\frac{1}{2}(N+X)\right]!\right) - \ln\left(\left[\frac{1}{2}(N-X)\right]!\right).$$

Applying Stirling's approximation to each term,

$$N \ln(N) - N - \frac{1}{2}(N+X) \ln\left(\frac{1}{2}(N+X)\right) + \frac{1}{2}(N+X) - \frac{1}{2}(N-X) \ln\left(\frac{1}{2}(N-X)\right) + \frac{1}{2}(N-X).$$

Then cleaning up the new terms a bit,

$$N \ln(N) - \mathcal{N} - \frac{1}{2} \left[(N+X) \ln\left(\frac{1}{2}(N+X)\right) + (N-X) \ln\left(\frac{1}{2}(N-X)\right) \right] + \mathcal{N}.$$

Now from looking at the next question we'll want to Taylor expand something like $\ln(1+x)$, so divide through the $\ln$ arguments by N ,

$$N \ln(N) - \frac{N}{2} \left[\left(1 + \frac{X}{N}\right) \ln\left(\frac{N}{2} \left(1 + \frac{X}{N}\right)\right) + \left(1 - \frac{X}{N}\right) \ln\left(\frac{N}{2} \left(1 - \frac{X}{N}\right)\right) \right]$$

then pull out the $N/2$ s into their own $\ln$ terms,

$$N \ln(N) - \frac{N}{2} \left[\left(1 + \frac{X}{N}\right) \ln\left(1 + \frac{X}{N}\right) + \left(1 - \frac{X}{N}\right) \ln\left(1 - \frac{X}{N}\right) + \left(1 + \frac{X}{N}\right) \ln\left(\frac{N}{2}\right) + \left(1 - \frac{X}{N}\right) \ln\left(\frac{N}{2}\right) \right]$$

and simplify

$$N \ln(N) - \frac{N}{2} \left[\left(1 + \frac{X}{N}\right) \ln\left(1 + \frac{X}{N}\right) + \left(1 - \frac{X}{N}\right) \ln\left(1 - \frac{X}{N}\right) + 2 \ln\left(\frac{N}{2}\right) \right].$$

Then we can pull the $2 \ln(N/2)$ out of the square brackets and get $-N \ln(N/2) = N \ln(2/N) = N(\ln(2) - \ln(N))$ so that the two $N \ln(N)$ terms cancel,

$$-\frac{N}{2} \left[\left(1 + \frac{X}{N}\right) \ln\left(1 + \frac{X}{N}\right) + \left(1 - \frac{X}{N}\right) \ln\left(1 - \frac{X}{N}\right) \right] + N \ln(2).$$

Now in the next bit we Taylor expand the terms that look like $(1 \pm x) \ln(1 \pm x)$.

(c) For $f(x) = \ln(1+x)$, $f'(x) = \frac{1}{1+x}$, $f''(x) = -\frac{1}{(1+x)^2}$, $f'''(x) = \frac{2}{(1+x)^3}$. Then, about zero,

$$f(x) \approx x - \frac{x^2}{2} + \frac{x^3}{3}.$$

(d) As mentioned in (b) we have terms like $(1 \pm x) \ln(1 \pm x)$. To second order these look like

$$(1+x) \ln(1+x) \approx (1+x) \left(x - \frac{x^2}{2} \right) \approx x + \frac{x^2}{2}$$

and

$$(1-x) \ln(1-x) \approx (1-x) \left(-x - \frac{x^2}{2} \right) \approx -x + \frac{x^2}{2}.$$

Then we can apply the expansion,

$$\left(1 + \frac{X}{N}\right) \ln\left(1 + \frac{X}{N}\right) + \left(1 - \frac{X}{N}\right) \ln\left(1 - \frac{X}{N}\right) \approx \frac{X}{N} + \frac{1}{2} \left[\frac{X}{N}\right]^2 - \frac{X}{N} + \frac{1}{2} \left[\frac{X}{N}\right]^2 = \left[\frac{X}{N}\right]^2.$$

Putting this back into the full expression then

$$\ln(\Omega(N, X)) \approx -\frac{N}{2} \left[\left[\frac{X}{N}\right]^2 \right] + N \ln(2) = N \ln(2) - \frac{X^2}{2N}.$$

(e) Exponentiating both sides we immediately obtain

$$\Omega \approx \exp\left(N \ln(2) - \frac{X^2}{2N}\right) = 2^N \exp\left(-\frac{X^2}{2N}\right).$$

This is a Gaussian, as we'd expect.